



ARCHER2

SP Quarterly Report

April - June 2026

EPCC

The University of Edinburgh



Document Information and Version History

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Reviewer(s)	Andy Turner, Jo Beech-Brandt

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0.1	01/07/2026	Template created	Jo Beech-Brandt
0.2	04/07/2026	Added narrative, graphs, service statistics	Jo Beech-Brandt
0.3	05/07/2026	Added critical success metrics	Lorna Smith
0.4	08/07/2026	Added narrative	Paul Clark
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1.0	13/07/2026	Version for UKRI	Jo Beech-Brandt, Andy Turner

1 The ARCHER2 Service

This is the report for the ARCHER2 SP Service for the Reporting Period: 1 April – 30 June 2026.

1.1 Service Highlights

- Utilisation remains high at 91% over the reporting period. The monthly utilisation figures were 96% in April, 90% in May and 88% in June. There were some cooling issues in the warmer months and capability days in May which have impacted utilisation slightly.
- EPCC successfully passed all of our annual ISO external audits in June. An external auditor ensures that EPCC continues to apply best practice and complies with the ISO 9001 (quality service delivery), ISO 27001 (information security) and ISO 22301 (business continuity and disaster recovery) standards. The considerable time and effort EPCC spend correlates with the importance we place on ensuring we run the best service for our users and keep user data and services as safe as possible whilst minimising service disruption
- EPCC is continuing work alongside EPSRC, NERC and HPE for the exit plans for the ARCHER2 service. The end date of the ARCHER2 service is 21st November 2026. Users have been notified of the end of service and a webpage advertised detailing initial information for transition. Further details will be added as the end of service approaches.
- ARCHER2 Staff attended and contributed to ISC in Hamburg in June where EPCC had a stand and took part in numerous workshops, tutorial and talks within the programme. We were able to meet with collaborators and promote HPC within the UK.
- Our summer engineering internship programme at the ACF started on the 8th June and our two interns are working to collate data and build models around our Building Management System to better support our aspirations around digital twin modelling.
- Our early career programme within the systems team will also see 2 x third-year undergraduate computing students starting on a one year placement, along with 2 x Graduate Apprenticeship (GA) students starting on a two year placement in early August. These roles will support the National services we run, along with our large-scale supercomputing data centre the ACF.
- New software images were successfully deployed containing patches for a number of recently emerged software vulnerabilities. These were deployed using a rolling reboot of compute nodes which limited user impact.
- Initial deployment of “View for Clusterstor” has been completed, and is now available to HPE to support investigations of file system issues.
- The annual ARCHER2 User Survey is currently open for user feedback and will remain open until September. Once it closes an analysis and report will be provided.

1.2 Forward Look

- With the unprecedented heatwave in the UK in the past 4-6 weeks, considerable work has been undertaken to help review the existing cooling infrastructure setup which supports ARCHER2 with a view to mitigate risks to the service on the hottest days. This work is ongoing and involves inputs from HPE, University of Edinburgh Estates team and an external contractor.
- Computer Room 4 (cr4) and Plant Room D (prD) have formally been handed over to our main contractors and work is progressing on the construction programme for the expanded room and underpinning infrastructure to host the Next National Supercomputing Service (NNSS). We expect significant visually impacting work to be complete over the next 2 quarters including the foundations and steelworks.
- Following the Capability Days during May, there will be additional capability days over the remainder of the lifetime of the ARCHER2 service. This will allow users the opportunity to test and run codes at large scale before the end of the service.
- Work continues towards ISO 14001 certification, the international standard for environmental management. This provides a framework for organisations to manage their environmental responsibilities, reduce impact (for example, waste, energy use, pollutants), ensure legal compliance, and continuously improve their environmental performance. This is particularly relevant for the ACF datacentre where we want to ensure we minimise the potential environmental impact of running such a resource intensive site.
- EPCC are continuing to work to integrate file system awareness into the Slurm scheduler, in order to allow individual file systems to be taken down for maintenance with less impact to users. This should reduce the overall unavailability for users of specific file systems to only the specific time where maintenance is conducted.

2 ARCHER2 Performance Report

This is the contractual performance report for the ARCHER2 SP Service for the Reporting Periods from 1 April until 30 June 2026.

2.1 Service Points and Service Credits

The Service Levels and Service Points for the SP service are defined by EPSRC in Schedule 2.2 of ARCHER2 SP Service Contract.

The Working Day (WD) for the ARCHER2 Service is 10 Working Hours (WH) as the Service operates from 0800-1800. The Median Time to Resolution is measured in WD.

- **Availability:** *Service Threshold: $\leq 96.5\%$; Operating Service Level: $>98.0\%$, $\leq 98.5\%$.*
- **ARCHER2_SP_Level1 (MTR):** The Median Time to Resolution, of all SP queries falling within Level 1 resolved by the Contractor in the Reporting Period. *MTR Service Threshold: >1 WD; Operating Service Level: >0.3 WD, ≤ 0.45 WD.*
- **ARCHER2_SP_Level2 (MTR):** The Median Time to Resolution, of all SP queries falling within Level 2 resolved by the Contractor in the Reporting Period. *MTR Service Threshold: >8 WD; Operating Service Level: >2 WD, ≤ 4 WD.*
- **ARCHER2_SP_Level3 (MTR):** The Median Time to Resolution, of all SP queries falling within Level 3 resolved by the Contractor in the Reporting Period. *MTR Service Threshold: >25 WD; Operating Service Level: >12 WD, ≤ 16 WD.*
- **Initial Response to Queries (%):** The percentage of the total number of SP queries assigned to the Contractor in the Reporting Period responded to within 3 Working Hours. *Service Threshold: $<96.00\%$; Operating Service Level: $98.00 - 98.99\%$.*
- **Query User Satisfaction (%):** The percentage of the total number of query satisfaction surveys completed in each Reporting Period, rating the quality of the resolution of Queries by the Contractor as “Good”, “Very Good” or “Excellent”. *Operating Service Level: $82.00 - 87.99\%$*

2.1.1 Service Points

Metric	Apr 2026		May 2026		Jun 2026		Q2 2026	
	Perf	Points	Perf	Points	Perf	Points	Perf	Points
Availability	100%	-3	100%	-3	100%	-3	100%	-9
SP_Level1 (MTR)	0.00	-2	0.00	-2	0.00	-2	0.00	-6
SP_Level2 (MTR)	0.05	-2	0.09	-2	0.09	-2	0.06	-6
SP_Level3 (MTR)	0.00	-2	8.00	-2	0.00	-2	8.00	-6
Initial Response (%)	100%	-1	100%	-1	100%	-1	100%	-3
Query Satisfaction (%)	100%	-2	100%	-2	100%	-2	100%	-6
Total		-12		-12		-12		-36

2.1.2 Service Credits

As the Total Service Points are negative (-36.0), no Service Credits apply in 2Q26.

2.2 SP Query Statistics

The metrics were specified by EPSRC in Schedule 2.2 of ARCHER2 SP Service Contract.

- **Assigned:** The number of SP queries assigned to the Contractor within each query resolution category in the Reporting Period.
- **Resolved:** The number of SP queries resolved by the Contractor within each query resolution category in the Reporting Period.
- **Backlog:** The number of SP queries assigned to the Contractor that remained unsolved within each query resolution category in the Reporting Period
- **Correspondence:** The average number of pieces of correspondence generated for SP queries in each query resolution category.
- **First Response:** The average time taken for the Contractor to first respond to the Originator of the SP query.

April 2026					
Service level	Assigned	Resolved	Backlog	Correspondence	First Response
SP_Level1	1871	1871	0	0.123	0:12:08
SP_Level2	65	70	14	6.686	0:14:08
SP_Level3	0	0	0	0	0:00:00
May 2026					
Service level	Assigned	Resolved	Backlog	Correspondence	First Response
SP_Level1	1314	1311	3	0.147	0:01:33
SP_Level2	54	52	16	7.827	0:19:34
SP_Level3	1	1	0	7	0:00:00
June 2026					
Service level	Assigned	Resolved	Backlog	Correspondence	First Response
SP_Level1	1739	1742	0	0.15	0:00:33
SP_Level2	92	69	39	7.159	2:29:56
SP_Level3	1	0	1	0	0:00:00
Q2 2026					
Service level	Assigned	Resolved	Backlog	Correspondence	First Response
SP_Level1	4924	4924	0	0.139	0:05:09
SP_Level2	211	191	39	7.168	1:03:45
SP_Level3	2	1	1	7	0:00:00

2.3 Query Resolution

Metric	Apr 2026		May 2026		Jun 2026		Q2 2026	
Service Level	MTR	Resolved	MTR	Resolved	MTR	Resolved	MTR	Resolved
SP_Level1	0:00:12	1871	0:00:05	1311	0:00:07	1742	0:00:08	4924
SP_Level2	0:32:00	70	0:54:05	52	0:54:28	69	0:37:32	191
SP_Level3	0:00:00	0	80:00:00	1	0:00:00	0	80:00:00	1
Total		1941		1364		1811		5116

A total of 5116 queries were resolved by the ARCHER2 SP Service in the Reporting Period. The percentage of user queries responded to within 3 hours was 100%.

2.4 Query Feedback

During April, there were 21 feedback scores received. 100% were Good, Very Good or Excellent with 95% given the highest score of Excellent.

During May, there were 9 feedback scores received. 100% were Good, Very Good or Excellent with 89% given the highest score of Excellent.

During June, there were 15 feedback scores received. 100% were Good, Very Good or Excellent with 93% given the highest score of Excellent.

An £45 donation was made to our chosen charity Save the Children with £1 donated per query feedback item received.

2.5 Maintenance and Outages

This quarter saw multiple partial outages due to extreme weather and the impact on the cooling system. Users could still connect to the login nodes, access their data and submit jobs during these incidents.

Type	Start	End	Duration	User Impact	Reason	Attributable
Partial	2026-04-23 1110	2026-04-23 1210	60 mins	University network issue has led to loss if DNS server and network connectivity. Login to ARCHER2 and SAFE may fail.	University-wide network issue.	Accommodation
Partial	2026-04-30 1550	2026-04-30 1800	2 hrs 10mins	New work was prevented from starting to reduce load on the cooling system. Users can login, access data and submit jobs.	Issue with cooling pump	Accommodation
Partial	2026-05-01 1320	2026-04-30 1405	45mins	New work was prevented from starting to reduce load on the cooling system. Users can login, access data	Issue with cooling pump	Accommodation

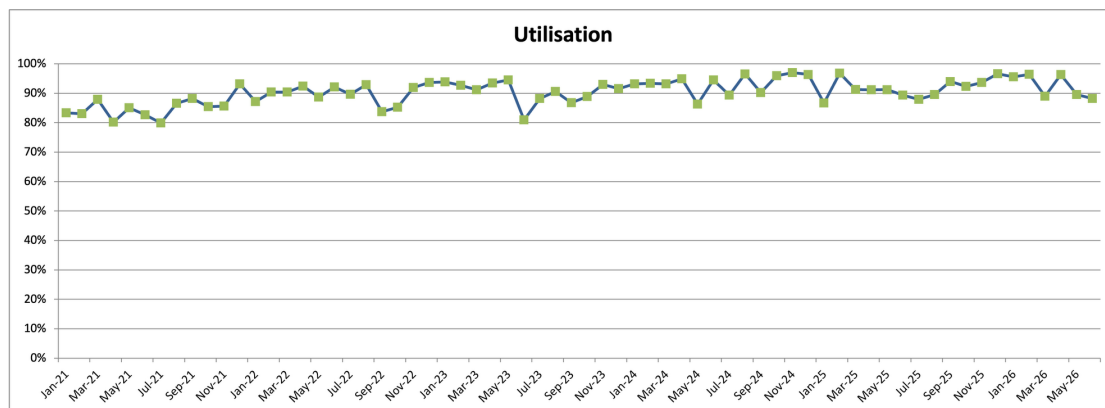
				and submit jobs.		
Partial	2026-04-30 1550	2026-04-30 1800	2 hrs 10mins	New work was prevented from starting to reduce load on the cooling system. Users can login, access data and submit jobs.	Issue with cooling pump	Accommodation
Partial	2026-05-25 0830	2026-05-27 1100	2 days 1hours 30mins	Reduced ARCHER2 capacity. Users can login, access data and submit jobs.	Issues with cooling infrastructure	Accommodation
Partial	2026-05-27 1100	2026-05-29 1000	2 days 1hours	Reduced ARCHER2 capacity. Users can login, access data and submit jobs.	Issues with cooling infrastructure	Accommodation
Partial	2026-05-29 1112	2026-05-29 1149	37mins	New jobs will not start. Users can login, access data and submit jobs.	Issues with cooling infrastructure	Accommodation
Partial	2026-05-25 0830	2026-05-27 1100	2 days 1hours 30mins	Reduced ARCHER2 capacity. Users can login, access data and submit jobs.	Issues with cooling infrastructure	Accommodation
Partial	2026-06-04 0830	2026-06-08 0130	3 days 17 hours	No real user impact other than slightly more wait time and possibly interruption to login node	All login and compute nodes to be rebooted to apply critical security fix	HPE
Partial	2026-06-15 1500	2026-06-16 1800	1 day 3 hours	No real user impact other than slightly more wait time	All compute nodes to be rebooted to	HPE

					apply critical security fix	
Partial	2026-06-17 1530	2026-06-17 2000	4hours 30 mins	New work prevented from starting. Reduced ARCHER2 capacity. Users can login, access data and submit jobs.	Issues with cooling infrastructure	Accommodation
Partial	2026-06-22 1200	2026-06-29 0000	6 days 12hours	Reduced ARCHER2 capacity. Users can login, access data and submit jobs.	Issues with cooling infrastructure	Accommodation

3 ARCHER2 Service Statistics

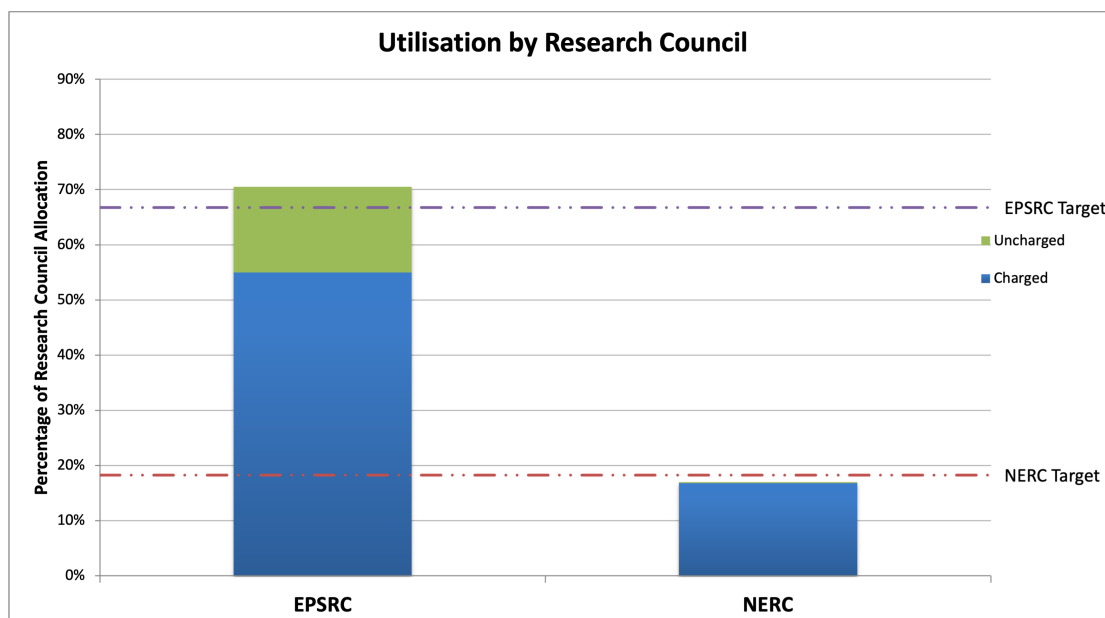
3.1 Utilisation

The utilisation for ARCHER2 remains high for 1 April – 30 June at 91% which is slightly down from last quarter. Utilisation for April was 96%, for May 90% and for June 88%. There were some cooling issues during June which can impact the utilisation.

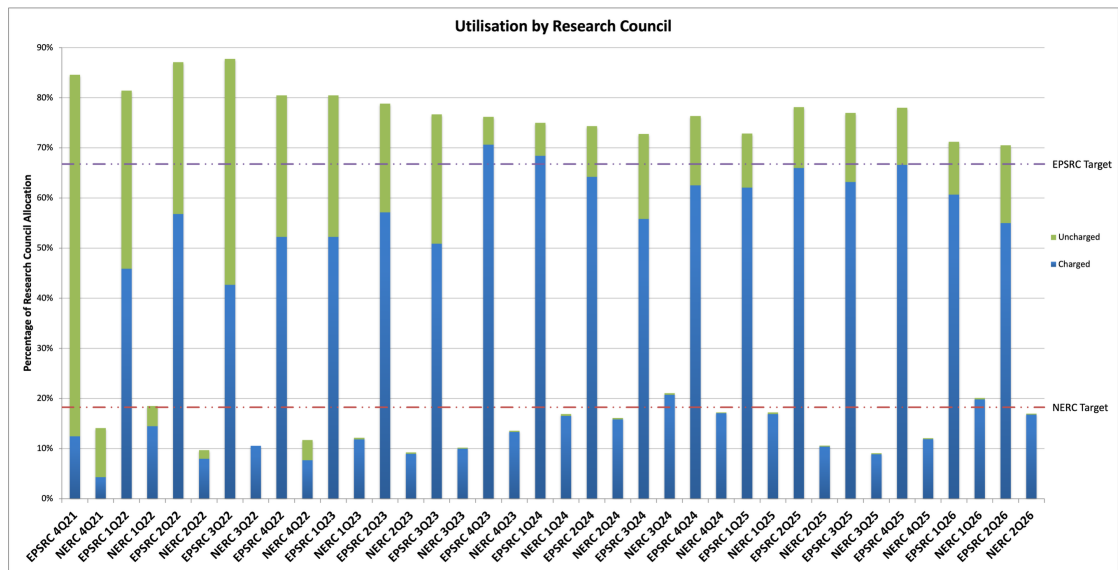


The utilisation by the Research Councils, relative to their respective allocations, is presented below. This bar chart shows the usage of ARCHER2 by the two Research Councils presented as a percentage of the total Research Council allocation on ARCHER2. It can be seen that EPSRC exceeded their target this quarter with their usage being at 71% (against their target of 66.8%).

NERC narrowly missed their target utilisation of 18.2% with their utilisation being 17% which is a decrease from the previous quarter of 20.1%.

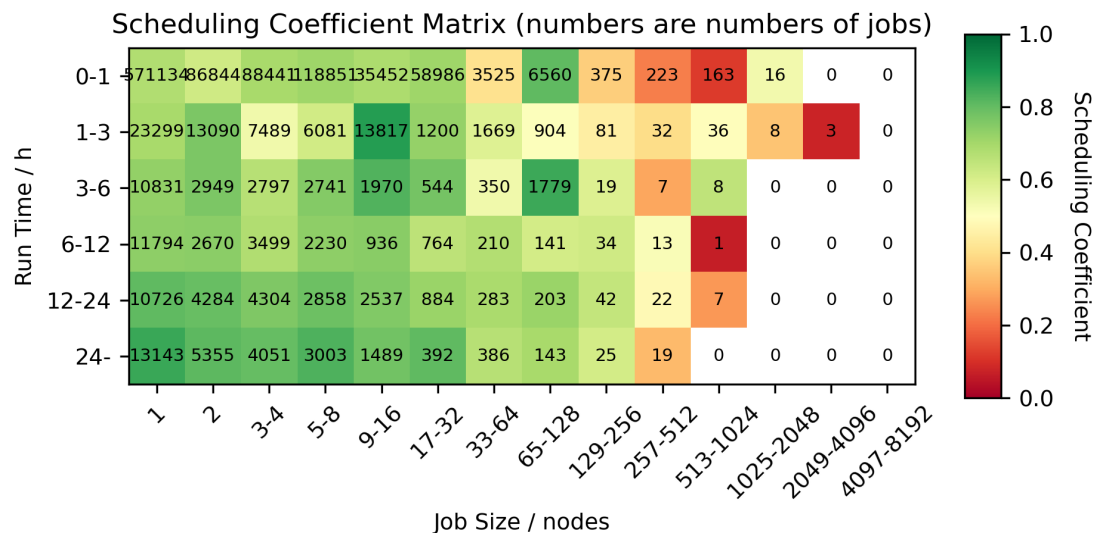


The stacked graph below shows the trend of charged and uncharged utilisation since the start of the service.

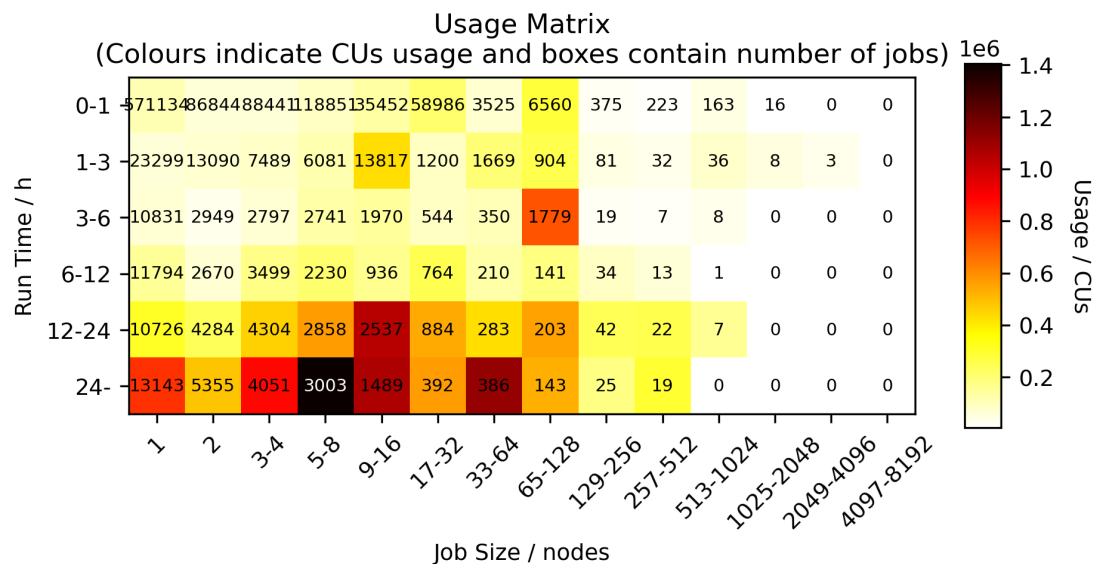


3.2 Scheduling Coefficient Matrix

The colour in the matrix indicates the value of the Scheduling Coefficient. This is defined as the ratio of runtime to runtime plus wait time. Hence, a value of 1 (green) indicates that a job ran with no time waiting in the queue, a value of 0.5 (pale yellow) indicates a job queued for the same amount of time that it ran, and anything below 0.5 (orange to red) indicates that a job queued for longer than it ran.



The usage heatmap below provides an overview of the usage on ARCHER2 over the quarter for different job sizes/lengths. The colour in the heatmap indicates the number of CUs expended for each class, and the number in the box is the number of jobs of that class.



Appendix: Critical Success Factors

1. Context

EPCC have been asked by UKRI to provide quarterly data for a number of critical success factors:

- CSF04 Implementation of environmentally considerate energy policies
- CSF07 Deliver and maintain a reliable data I/O function
- CSF08 Be cost-effective, cost-efficient and drive towards lowering of operational costs

In the sections below, please find the relevant metrics and data.

2. CSF04 Implementation of environmentally considerate energy policies

Implementation of environmentally considerate energy policies with a drive to reducing costs and environmental impacts.

All electricity provided to the ACF and ARCHER2 is on a 100% green, renewable energy tariff.

Environmentally considerate policies: 4

Since the start of full Service, EPCC have worked on implementing the following policies:

- Move from High Performance Mode to Low Power Mode: reduced average power draw from 3.2 MW to 2.9 MW (9%) with negligible impact on performance [May 2022]
- Reduced default processor frequency: further reduced average power to around 2.5 MW (19%) [December 2022]
- Increase in coolant temperatures: this will result in an increase in passive cooling (“free cooling”) [ongoing]
- Developed a set of new tools to help users estimate the environmental impact of their computing simulations and workloads [November 2024]

Power Usage

	4Q 21	1Q 22	2Q 22	3Q 22	4Q 22	1Q 23	2Q 23	3Q 23	4Q 23	1Q24
Average Power	3.31	3.16	3.15	2.86	2.90	2.51	2.56	2.46	2.53	2.58
	1Q 24	2Q 24	3Q 24	4Q 24	1Q 25	2Q 25	3Q 25	4Q25	1Q26	2Q26
Average Power	2.58	2.54	2.64	2.57	2.55	2.55	2.52	2.54	2.53	2.49

4Q 21 is Partial

3 & 4Q 25 does not include period of site downtime

So far, the average power draw has been reduced by around 0.7MW (21%) which will reduce electricity usage by up to 6M kWh per annum, significantly reducing annual running costs.

3. CSF07 Deliver and maintain a reliable data I/O function

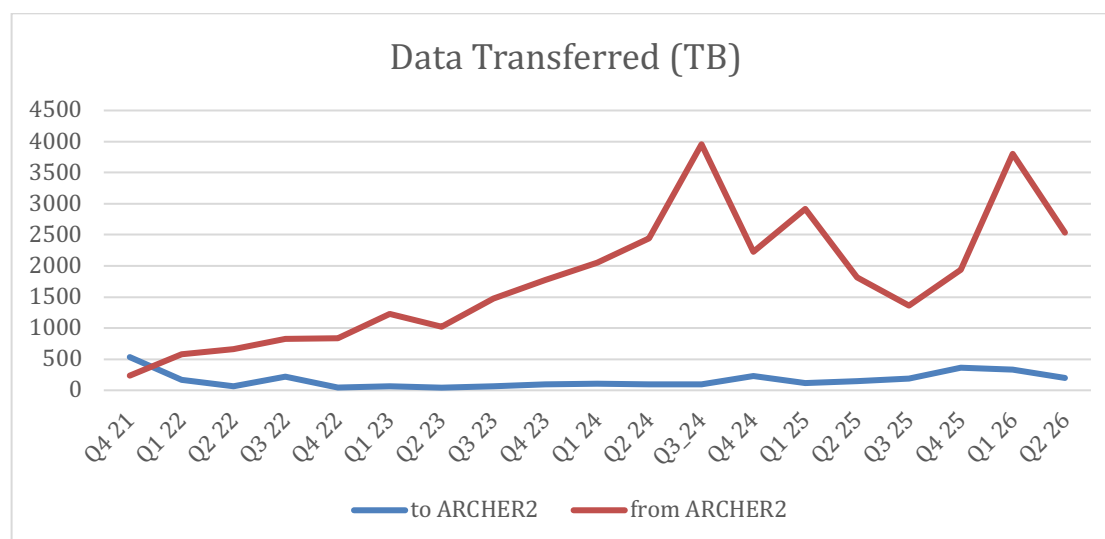
The compute resource will deliver and maintain an efficient, effective and reliable data I/O function which meets the requirements of users and their software. It will evolve and expand to accommodate new software or hardware architectures as required by the Service or its user base.

Data Transferred

EPCC monitor the data transfer rates in and out of the ARCHER2 system. Based on this, we now estimate the total amount of data transferred on and off ARCHER2 each Quarter.

Data Transferred...	4Q21*	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23
...to ARCHER2 (TB)	534	163	68	220	44	67	42	65	99
...from ARCHER2(TB)	236	582	667	822	834	1231	1022	1472	1771
Data Transferred...	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26
...to ARCHER2 (TB)	108	93	98	228	114	150	191	361	335
...from ARCHER2(TB)	2056	2443	3956	2227	2915	1815	1359	1934	3802
Data Transferred...	2Q26								
...to ARCHER2 (TB)	201								
...from ARCHER2(TB)	2541								

* Partial



The amount of data moved off ARCHER2 has been in line with overall trends, although down on the previous quarter as last year.

Parallel IO write performance

We regularly monitor the parallel write performance between the compute nodes and the parallel Lustre (/work) file systems. We use the benchio synthetic IO benchmark application¹ and report the MPI-IO write performance with the following settings:

- Global data structure of 2048³: writes a single file of 65,536 MiB (64 GiB).
- Uses 16 compute nodes and 128 MPI processes per node.
- Uses UCX as the MPI transport protocol.
- Sets the following environment variables:
 - FI_OFI_RXM_SAR_LIMIT=64K
 - MPICH_MPIIO_HINTS="*:cray_cb_write_lock_mode=2,*:cray_cb_nodes_multiplier=4"

These settings have been found to maximise the IO performance for parallel writes using MPI-IO on the ARCHER2 file systems. Writes using the default settings on ARCHER2 typically have median write values 2-3 GiB/s lower than the optimised values.

Original reporting of this data (Q1 and Q2 2023) used the means from a small number of runs on the HDD-based Lustre file systems. From Q3 2023 onwards we have been monitoring performance regularly on both HDD and NVMe-based Lustre file systems throughout the quarter and report median (Q2) and lower (Q1) and upper quartile (Q3) performance and provide boxplots illustrating the performance variation. (On the boxplots, the green triangles mark the mean value and the whiskers extend to the last datapoint within the range 1.5 x IQR.)

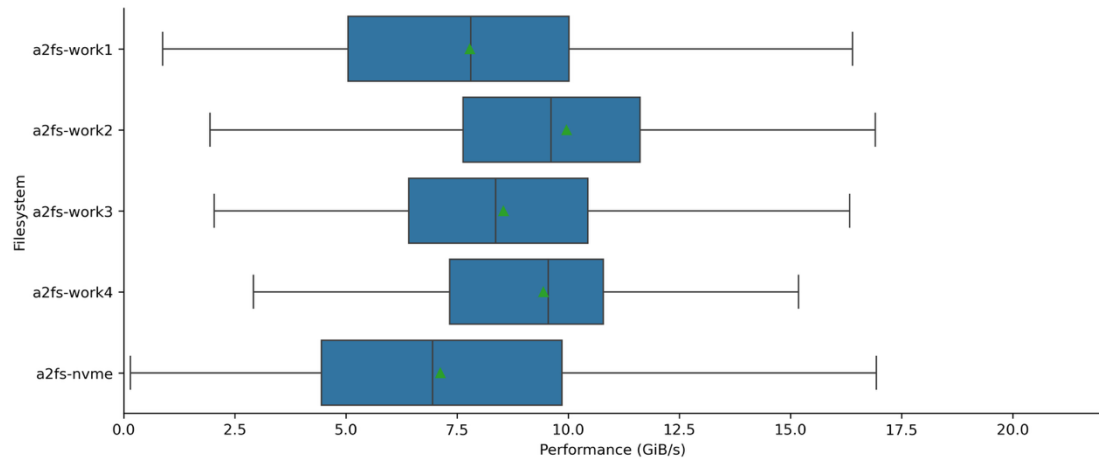
During Q1 2025, we have worked to remove data from a2fs-work1 as much as possible. With the usage of the file system now below 80% we see a significant improvement in performance compared to Q4 2024 where the usage on the file system was over 80%. We are working to ensure that usage on any of the ARCHER2 Lustre file systems does not go above 80% to try to maintain good performance for users on the service.

Lower performance of a2fs-work1 during 4Q 2024 and 4Q 2025 are both attributed to usage levels moving close to or beyond 80% of total file system capacity leading to less efficient data storage. Similarly, the wider spread of performance for a2fs-nvme (the solid state, scratch) file system during Q2 2026 was also due to usage levels moving beyond 80% of total file system capacity during this period.

¹ <https://github.com/davidhenty/benchio>

Benchio MPI-IO medium (GiB/s)	1Q23	2Q23		3Q23	4Q23	1Q24	2Q24	3Q24	4Q24
a2fs- work1	8.2	7.6±0.5		10.5 (8.8:11.8)	10.9 (8.3:12.5)		10.1 (7.0:11.8)	9.7 (6.7:11.9)	4.0 (2.0:9.6)
a2fs- work2	8.5	7.3±0.6		10.4 (7.2:12.4)	10.4 (7.7:13.0)		11.1 (8.0:12.5)	11.1 (8.1:13.1)	11.3 (7.1:15.0)
a2fs- work3	8.3	9.6±0.7		10.0 (8.2:11.6)	10.7 (8.1:11.9)		9.6 (8.4:11.8)	9.6 (7.5:11.8)	9.3 (7.0:11.6)
a2fs- work4					9.7 (9.1:10.2)		10.0 (9.2:10.8)	10.6 (9.4:11.6)	10.7 (9.5:11.7)
a2fs- nvme				10.1 (9.6:11.5)	10.1 (9.5:12.4)		11.1 (10.5:12.4)	11.6 (11.1:12.7)	10.7 (10.0:11.8)

Benchio MPI-IO medium (GiB/s)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26
a2fs- work1	8.6 (5.0:11.1)	8.6 (6.4:11.7)	7.5 (4.9:9.7)	3.9 (2.0:6.8)	6.5 (4.5:8.5)	7.8 (5.1:10.0)
a2fs- work2	10.7 (8.1:12.7)	11.4 (8.9:12.6)	10.3 (8.1:12.0)	8.6 (7.0:10.9)	8.6 (6.9:10.7)	9.6 (7.6:11.6)
a2fs- work3	9.3 (7.7:11.6)	9.5 (7.3:11.7)	8.8 (7.2:10.6)	7.6 (6.1:8.8)	7.5 (6.3:8.6)	8.4 (6.4:10.4)
a2fs- work4	10.1 (8.8:10.9)	10.5 (9.0:11.3)	9.5 (7.8:10.5)	7.8 (6.8:8.6)	7.9 (6.8:8.9)	9.6 (7.3:10.8)
a2fs- nvme	11.3 (10.4:13.3)	11.0 (10.4:13.5)	10.9 (10.4:13.1)	8.3 (7.9:9.6)	7.6 (6.7:8.4)	7.0 (4.5:9.9)



4. CSF08 Be cost-effective, cost-efficient and drive towards lowering of operational costs

The Service shall be cost-effective and cost-efficient across its elements during its lifetime and drive towards lowering of operational costs by seeking efficiencies in delivery such that TCO presents an acceptable and cost-effective solution for the public. The Service will monitor and report its Power Usage Effectiveness (PUE) and strive to make efficiency savings where possible.

Relative Research Output

Measure	11/2021 – /2022	5/2022 – 12/2022	01/2023 – 12/2023	1Q 24	2Q 24	3Q 24	4Q 24
Relative Research Output per kWh	100	109	115	115	115	115	115
Measure	1Q 25	2Q 25	3Q 25	4Q25	1Q26	2Q25	
Relative Research Output per kWh	115	115	115	115	115	115	

We define the initial measure of research output per kWh on ARCHER2 to be 100 and then estimate how this has changed with the introduction of the various environmentally considerate policies discussed under CSF04. This is estimated using applications benchmarks similar to those defined by UKRI for the procurement.

Energy Used per CU Delivered

	4Q21*	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26
Energy per CU (kWh)	0.719	0.713	0.728	0.715	0.650	0.545	0.669	0.590	0.568	0.582	0.585	0.595	0.546	0.518	0.578	0.638	0.602	0.615

Energy Cost per CU Delivered

	4Q21*	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26
Cost per CU (£)	£0.089	£0.090	£0.098	£0.096	£0.088	£0.074	£0.162	£0.143	£0.136	£0.140	£0.160	£0.164	£0.149	£0.142	£0.131	£0.139	£0.130	£0.133

* Partial

The two tables above are calculated using the total CUs delivered by ARCHER2, the total kWh of electricity consumed, and the unit cost for kWh. The increase in “Energy Cost per CU Delivered” from 2Q23 is caused by a significant increase in the unit cost of electricity from April 2023. For 2Q23, there is also an impact on the “Energy Used per CU Delivered” from the major software upgrade that took 3 weeks. There was also an additional increase in the unit cost of electricity from April 2024.